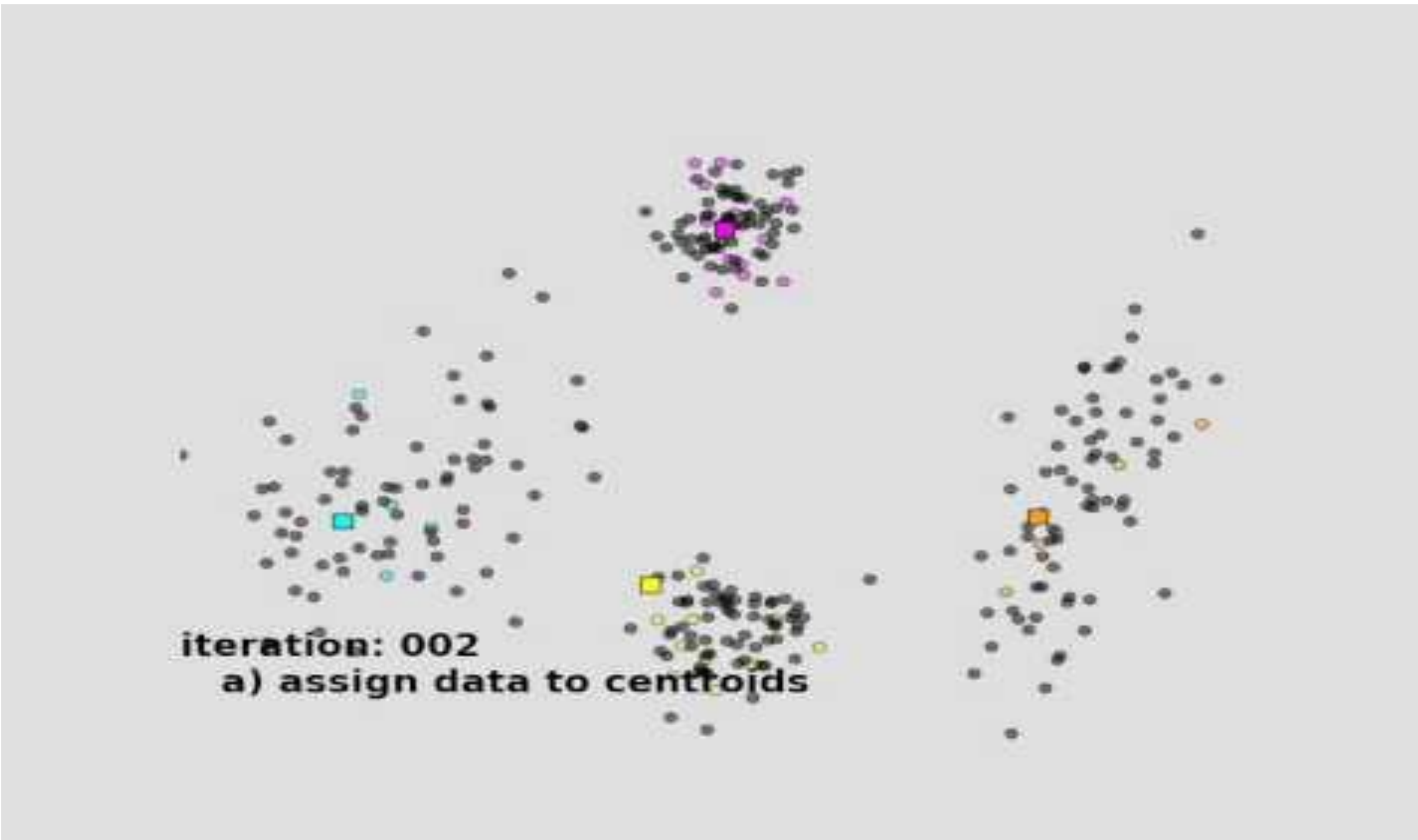
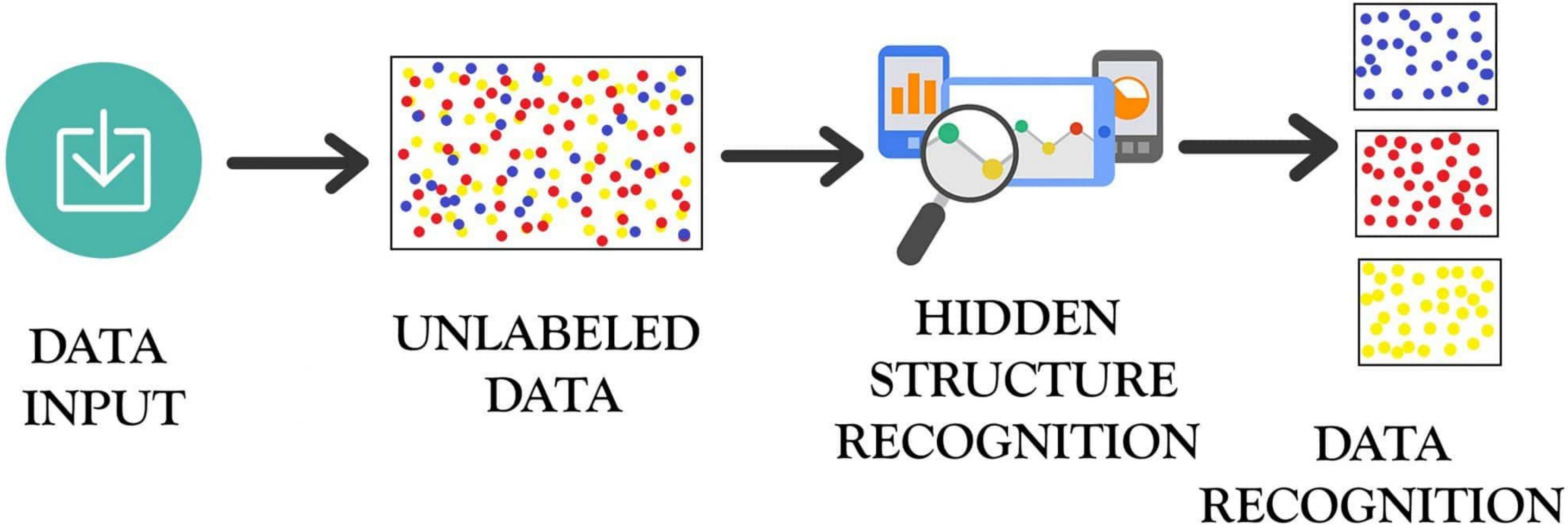
K-MEANS CLUSTERING


iteration: 002
a) assign data to centroids

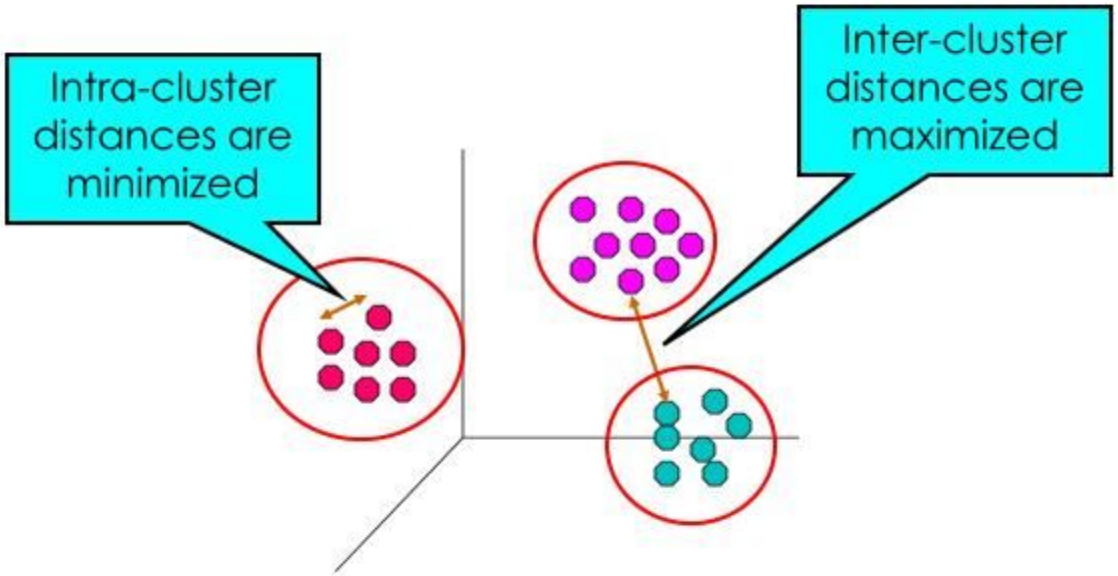
Machine learning works well if we have a labeled data. But what about the cases when we do not have the labeled data? One option is to get some labeled data and the other is to use unsupervised machine learning. Unsupervised ML algorithms discover the interesting patterns in data for learning without any label data and teacher for guidance.

UNSUPERVISIED LEARNING



unsupervised learning - clustering

- Finding groups of objects such that the objects in a group will be similar (or related) to one another and different from (or unrelated to) the objects in other groups



after finding the relationship patterns, clusters the data samples that have similar features into groups.

sample

Cluster/group

Unsupervised learning problems can be further grouped into clustering and association problems.

- Clustering: A clustering problem is where you want to discover the inherent groupings in the data, such as grouping customers by purchasing behavior.
- Association: An association rule learning problem is where you want to discover rules that describe large portions of your data, such as people that buy X also tend to buy Y.

What is Unsupervised Learning?

date	customer	account	auth	class	zip	amount
Mon	Bob	3421	pin	clothes	46140	135
Tue	Bob	3421	sign	food	46140	401
Tue	Alice	2456	pin	food	12222	234
Wed	Sally	6788	pin	gas	26339	94
Wed	Bob	3421	pin	tech	21350	2459
Wed	Bob	3421	pin	gas	46140	83
Thr	Sally	6788	sign	food	26339	51

What is Unsupervised Learning?

date	customer	account	auth	class	zip	amount
Mon	Bob	3421	pin	clothes	46140	135
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Wed	Bob	3421	pin	tech	21350	2459
Wed	Bob	3421	pin	gas	46140	83
The	Sally	6788	sign	food	26339	51

similar

INTRODUCTION- What is clustering?

- **Clustering** is the classification of objects into different groups, or more precisely, the partitioning of a data set into subsets (clusters), so that the data in each subset (ideally) share some common trait - often according to some defined distance measure.

Types of clustering:

1. **Hierarchical algorithms**: these find successive clusters using previously established clusters.

    1. Agglomerative ("bottom-up"): Agglomerative algorithms begin with each element as a separate cluster and merge them into successively larger clusters.
    2. Divisive ("top-down"): Divisive algorithms begin with the whole set and proceed to divide it into successively smaller clusters.

2. **Partitional clustering:** Partitional algorithms determine all clusters at once. They include:

- ***K*-means and derivatives**
- Fuzzy *c*-means clustering
- QT clustering algorithm

FUNDAMENTAL CONCEPT

Why Distance Metrics Matter

Distance Defines Similarity

distance is the primary mechanism for identifying similar data points. The choice of distance metric fundamentally determines which points are considered "nearest" and therefore influences every prediction the algorithm makes.

> "Different distance metrics create different notions of similarity. What is 'near' in Euclidean space may be 'far' in Manhattan space."

Impact on Decision Boundaries

Different distance metrics produce dramatically different decision boundaries. The same dataset can yield different predictions depending on which metric you choose.

Euclidean	Manhattan	Cosine
Circular boundaries	Diamond boundaries	Angular boundaries

Choosing the Right Metric

The optimal distance metric depends on your data characteristics, feature types, and problem domain. There is no universal "best" metric—only the most appropriate one for your specific use case.

Continuous Features: Euclidean, Manhattan, Minkowski

Text/NLP: Cosine similarity

Categorical: Hamming distance

Common Distance Metrics

Euclidean
L2 Norm
Straight-line distance in continuous space

Manhattan
L1 Norm
Grid-based movement (city blocks)

Minkowski
Generalized
Parameterized generalization

Cosine
Angular
Direction-based similarity

Hamming
Categorical
Different positions count

Key Insight

The distance metric is a hyperparameter that should be tuned alongside K. Experiment with multiple metrics and validate performance to find the optimal choice for your dataset.

CORE METRICS

Euclidean & Manhattan Distance

Euclidean Distance

L2 Norm · Straight-Line

The "ordinary" straight-line distance between two points in Euclidean space. It's the most intuitive and commonly used distance metric, derived from the Pythagorean theorem.

Formula (2D)

$$d = \sqrt{[(x_2-x_1)^2 + (y_2-y_1)^2]}$$

General Formula (n-D)

$$d = \sqrt{[\Sigma(x_i - y_i)^2]}$$

Best For

- Continuous, dense features
- Low-dimensional data
- Geometric interpretations

Limitations

- Sensitive to outliers
- Struggles in high dimensions

Geometric Interpretation: The shortest path between two points

Manhattan Distance

L1 Norm · City Block

Also known as "city block" or "taxicab" distance, Manhattan distance measures the distance between two points by only moving along axes (like navigating city blocks).

Formula (2D)

$$d = |x_2-x_1| + |y_2-y_1|$$

General Formula (n-D)

$$d = \Sigma|x_i - y_i|$$

Best For

- High-dimensional data
- Grid-based movement
- Robust to outliers

Limitations

- Less intuitive geometry
- Ignores diagonal relationships

Geometric Interpretation: Diamond-shaped distance contours

ADVANCED METRICS

Minkowski, Cosine & Hamming Distances

Minkowski

A generalized distance metric that encompasses both Euclidean and Manhattan as special cases through a parameter p.

Formula

$$d = (\Sigma|x_i - y_i|^p)^{(1/p)}$$

p = 1 → Manhattan
Grid-based distance

p = 2 → Euclidean
Straight-line distance

p → ∞ → Chebyshev
Maximum coordinate difference

Use Case: Tune p for optimal performance

Cosine

Measures the cosine of the angle between two vectors, focusing on direction rather than magnitude. Essential for high-dimensional sparse data.

Formula

$$\text{similarity} = (A \cdot B) / (||A|| \times ||B||)$$

Range: [-1, 1]
1 = same direction, -1 = opposite

Distance = 1 - Similarity
Convert to distance metric

Ignores Magnitude
Focuses purely on orientation

Use Case: Text mining, document similarity

Hamming

Counts the number of positions at which corresponding values differ. Designed for categorical and binary data.

Formula

$$d = \Sigma(x_i \neq y_i)$$

Binary Data
Count differing bits

Categorical Data
Count mismatched categories

Normalized Version
d = (mismatches) / (total)

Use Case: DNA sequences, error detection

K-MEANS CLUSTERING

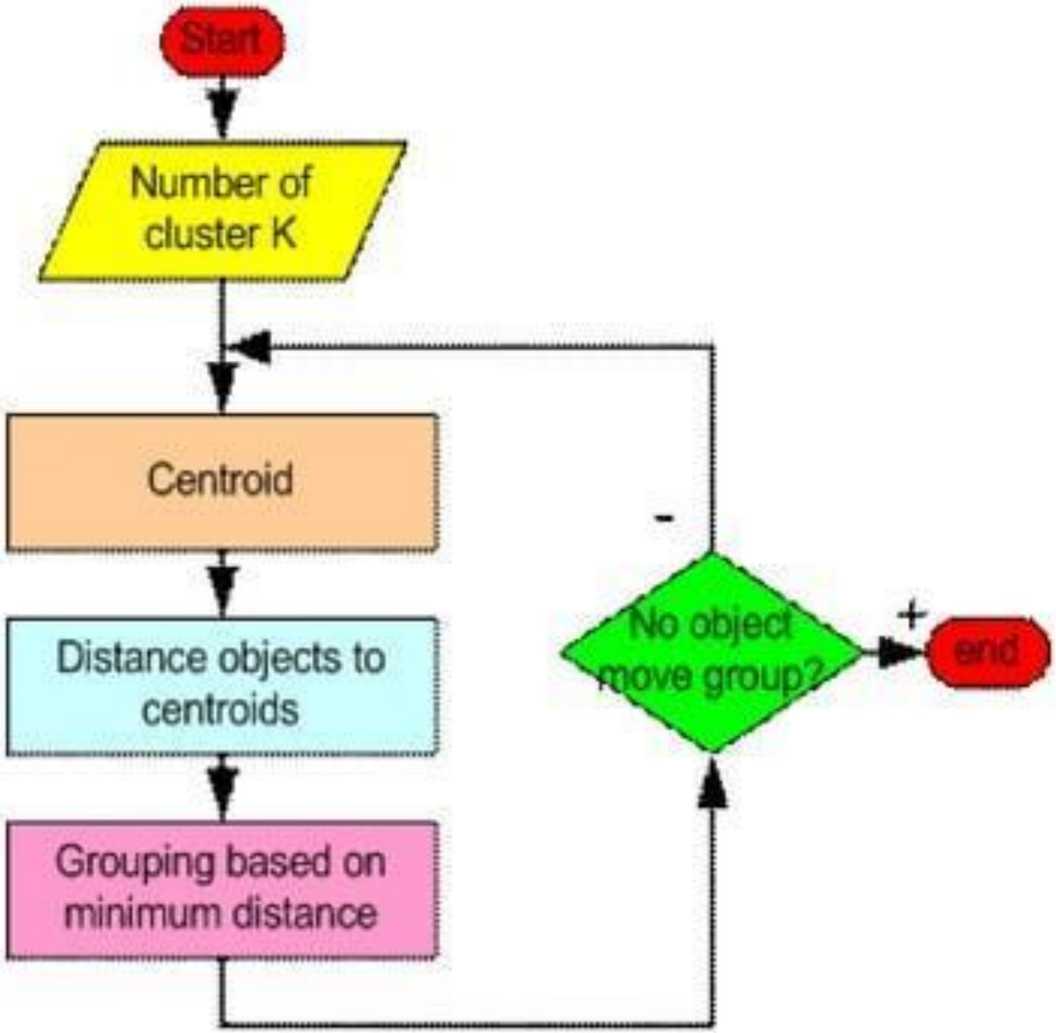
- The **k-means algorithm** is an algorithm to cluster *n* objects based on attributes into *k* partitions, where $k < n$.
- It is similar to the expectation-maximization algorithm for mixtures of Gaussians in that they both attempt to find the centers of natural clusters in the data.
- It assumes that the object attributes form a vector space.

- Simply speaking k-means clustering is an algorithm to classify or to group the objects based on attributes/features into K number of group.
- K is positive integer number.
- The grouping is done by minimizing the sum of squares of distances between data and the corresponding cluster centroid.

Working of K-means algorithm

1. Initialization: Choose the number of clusters (k) you want to create and randomly initialize k cluster centroids. Each centroid represents the center of a cluster.
2. Assignment Step: For each data point, calculate its distance from each of the k centroids. Assign the data point to the cluster whose centroid is closest to it. This forms k initial clusters.
3. Update Step: Recalculate the centroids of the k clusters by taking the mean of all the data points assigned to each cluster.
4. Repeat: Repeat the assignment and update steps iteratively until either a maximum number of iterations is reached or the centroids stop changing significantly.
5. Convergence: The algorithm is considered to have converged when the centroids of the clusters no longer change substantially between iterations.

How the K-Mean Clustering algorithm works?



Working of K-means algorithm: Example

Point	Coordinates
A1	(2,10)
A2	(2,6)
A3	(11,11)
A4	(6,9)
A5	(6,4)
A6	(1,2)
A7	(5,10)
A8	(4,9)
A9	(10,12)
A10	(7,5)
A11	(9,11)
A12	(4,6)
A13	(3,10)
A14	(3,8)
A15	(6,11)

STEP1: randomly initialize k cluster centroids

Centroid 1=(2,6) is associated with cluster 1.

Centroid 2=(5,10) is associated with cluster 2.

Centroid 3=(6,11) is associated with cluster 3.

Working of K-means algorithm: Example

Point	Distance from Centroid 1 (2,6)	Distance from Centroid 2 (5,10)	Distance from Centroid 3 (6,11)	Assigned Cluster
A1 (2,10)	4	3	4.123106	Cluster 2
A2 (2,6)	0	5	6.403124	Cluster 1
A3 (11,11)	10.29563	6.082763	5	Cluster 3
A4 (6,9)	5	1.414214	2	Cluster 2
A5 (6,4)	4.472136	6.082763	7	Cluster 1
A6 (1,2)	4.123106	8.944272	10.29563	Cluster 1
A7 (5,10)	5	0	1.414214	Cluster 2
A8 (4,9)	3.605551	1.414214	2.828427	Cluster 2
A9 (10,12)	10	5.385165	4.123106	Cluster 3
A10 (7,5)	5.09902	5.385165	6.082763	Cluster 1
A11 (9,11)	8.602325	4.123106	3	Cluster 3
A12 (4,6)	2	4.123106	5.385165	Cluster 1
A13 (3,10)	4.123106	2	3.162278	Cluster 2
A14 (3,8)	2.236068	2.828427	4.242641	Cluster 1
A15 (6,11)	6.403124	1.414214	0	Cluster 3

The distance function between two points
a = (x1, y1) and b = (x2, y2) is defined as-

Euclidean Distance =
$\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

$(x_1, y_1) = (2, 6)$
$(x_2, y_2) = (2, 10)$

Euclidean Distance C1

$$= \sqrt{(2 - 2)^2 + (6 - 10)^2}$$
$$= \sqrt{0^2 + 4^2}$$
$$= \sqrt{0 + 16}$$
$$= \sqrt{16}$$
$$= 4$$

Working of K-means algorithm: Example

Point	Distance from Centroid 1 (2,6)	Distance from Centroid 2 (5,10)	Distance from Centroid 3 (6,11)	Assigned Cluster
A1 (2,10)	4	3	4.123106	Cluster 2
A2 (2,6)	0	5	6.403124	Cluster 1
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A14 (3,8)	2.236068	2.828427	4.242641	Cluster 1
A15 (6,11)	6.403124	1.414214	0	Cluster 3

Euclidean Distance = $\sqrt{((x_2 - x_1)^2 + (y_2 - y_1)^2)}$

$(x_1, y_1) = (5, 10)$
$(x_2, y_2) = (2, 10)$

Euclidean Distance C2

$$= \sqrt{((5 - 2)^2 + (10 - 10)^2)}$$
$$= \sqrt{(3^2 + 0^2)}$$
$$= \sqrt{(9 + 0)}$$
$$= \sqrt{9}$$
$$= 3$$

Working of K-means algorithm: Example

Point	Distance from Centroid 1 (2,6)	Distance from Centroid 2 (5,10)	Distance from Centroid 3 (6,11)	Assigned Cluster
A1 (2,10)	4	3	4.123106	Cluster 2
A2 (2,6)	0	5	6.403124	Cluster 1
A3 (11,11)	10.29563	6.082763	5	Cluster 3
A4 (6,9)	5	1.414214	2	Cluster 2
A5 (6,4)	4.472136	6.082763	7	Cluster 1
A6 (1,2)	4.123106	8.944272	10.29563	Cluster 1
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A12 (4,6)	2	4.123106	5.385165	Cluster 1
A13 (3,10)	4.123106	2	3.162278	Cluster 2
A14 (3,8)	2.236068	2.828427	4.242641	Cluster 1
A15 (6,11)	6.403124	1.414214	0	Cluster 3

Euclidean Distance = $\sqrt{((x_2 - x_1)^2 + (y_2 - y_1)^2)}$

$(x_1, y_1) = (6, 11)$
$(x_2, y_2) = (2, 10)$

Euclidean Distance C3

$$= \sqrt{((6 - 2)^2 + (11 - 10)^2)}$$
$$= \sqrt{(4^2 + 1^2)}$$
$$= \sqrt{(16 + 1)}$$
$$= \sqrt{17}$$
$$= 4.12$$

Working of K-means algorithm: Example

Now, we will calculate the new centroid for each cluster.

In cluster 1, we have 6 points i.e. A2 (2,6), A5 (6,4), A6 (1,2), A10 (7,5), A12 (4,6), A14 (3,8).

To calculate the new centroid for cluster 1, we will find the mean of the x and y coordinates of each point in the cluster.

Mean= [(2+6+1+7+4+3) /6 + (6+4+2+5+6+8)/6]=(23/6) +(31/6)

Hence, the new centroid for cluster 1 is (3.833, 5.167).

Working of K-means algorithm: Example

In cluster 2, we have 5 points i.e. A1 (2,10), A4 (6,9), A7 (5,10) , A8 (4,9), and A13 (3,10).

Hence, the new centroid for cluster 2 is (4, 9.6)

In cluster 3, we have 4 points i.e. A3 (11,11), A9 (10,12), A11 (9,11), and A15 (6,11).

Hence, the new centroid for cluster 3 is (9, 11.25).

Working of K-means algorithm: Example

Point	Distance from Centroid 1 (3.833, 5.167)	Distance from centroid 2 (4, 9.6)	Distance from centroid 3 (9, 11.25)	Assigned Cluster
A1 (2,10)	5.169	2.040	7.111	Cluster 2
A2 (2,6)				
A3 (11,11)				
A4 (6,9)				
A5 (6,4)				
A6 (1,2)				
A7 (5,10)				
A8 (4,9)				
A9 (10,12)				
A10 (7,5)				
A11 (9,11)				
A12 (4,6)				
A13 (3,10)				
A14 (3,8)				
A15 (6,11)				

Working of K-means algorithm: Example

Point	Distance from Centroid 1 (3.833, 5.167)	Distance from centroid 2 (4, 9.6)	Distance from centroid 3 (9, 11.25)	Assigned Cluster
A1 (2,10)	5.169	2.040	7.111	Cluster 2
A2 (2,6)	2.013	4.118	8.750	Cluster 1
A3 (11,11)	9.241	7.139	2.016	Cluster 3
A4 (6,9)	4.403	2.088	3.750	Cluster 2
A5 (6,4)	2.461	5.946	7.846	Cluster 1
A6 (1,2)	4.249	8.171	12.230	Cluster 1
A7 (5,10)	4.972	1.077	4.191	Cluster 2
A8 (4,9)	3.837	0.600	5.483	Cluster 2
A9 (10,12)	9.204	6.462	1.250	Cluster 3
A10 (7,5)	3.171	5.492	6.562	Cluster 1
A11 (9,11)	7.792	5.192	0.250	Cluster 3
A12 (4,6)	0.850	3.600	7.250	Cluster 1
A13 (3,10)	4.904	1.077	6.129	Cluster 2
A14 (3,8)	2.953	1.887	6.824	Cluster 2
A15 (6,11)	6.223	2.441	3.010	Cluster 2

Working of K-means algorithm: Example

Now, we will calculate the new centroid for each cluster for the third iteration.

1. In cluster 1, we have 5 points i.e. A2 (2,6), A5 (6,4), A6 (1,2), A10 (7,5), and A12 (4,6). To calculate the new centroid for cluster 1, we will find the mean of the x and y coordinates of each point in the cluster. Hence, the new centroid for cluster 1 is
2. In cluster 2, we have 7 points i.e. A1 (2,10), A4 (6,9), A7 (5,10) , A8 (4,9), A13 (3,10), A14 (3,8), and A15 (6,11). Hence, the new centroid for cluster 2 is
3. In cluster 3, we have 3 points i.e. A3 (11,11), A9 (10,12), and A11 (9,11). Hence, the new centroid for cluster 3 is

Working of K-means algorithm: Example

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2. In cluster 2, we have 7 points i.e. A1 (2,10), A4 (6,9), A7 (5,10) , A8 (4,9), A13 (3,10), A14 (3,8), and A15 (6,11). Hence, the new centroid for cluster 2 is (4.143, 9.571)
3. In cluster 3, we have 3 points i.e. A3 (11,11), A9 (10,12), and A11 (9,11). Hence, the new centroid for cluster 3 is (10, 11.333).

Working of K-means algorithm: Example

Point	Distance from Centroid 1 (4, 4.6)	Distance from centroid 2 (4.143, 9.571)	Distance from centroid 3 (10, 11.333)	Assigned Cluster
A1 (2,10)	5.758	2.186	8.110	Cluster 2
A2 (2,6)				
A3 (11,11)				
A4 (6,9)				
A5 (6,4)				
A6 (1,2)				
A7 (5,10)				
A8 (4,9)				
A9 (10,12)				
A10 (7,5)				
A11 (9,11)				
A12 (4,6)				
A13 (3,10)				
A14 (3,8)				
A15 (6,11)				

Working of K-means algorithm: Example

Point	Distance from Centroid 1 (4, 4.6)	Distance from centroid 2 (4.143, 9.571)	Distance from centroid 3 (10, 11.333)	Assigned Cluster
A1 (2,10)	5.758	2.186	8.110	Cluster 2
A2 (2,6)	2.441	4.165	9.615	Cluster 1
A3 (11,11)	9.485	7.004	1.054	Cluster 3
A4 (6,9)	4.833	1.943	4.631	Cluster 2
A5 (6,4)	2.088	5.872	8.353	Cluster 1
A6 (1,2)	3.970	8.197	12.966	Cluster 1
A7 (5,10)	5.492	0.958	5.175	Cluster 2
A8 (4,9)	4.400	0.589	6.438	Cluster 2
A9 (10,12)	9.527	6.341	0.667	Cluster 3
A10 (7,5)	3.027	5.390	7.008	Cluster 1
A11 (9,11)	8.122	5.063	1.054	Cluster 3
A12 (4,6)	1.400	3.574	8.028	Cluster 1
A13 (3,10)	5.492	1.221	7.126	Cluster 2
A14 (3,8)	3.544	1.943	7.753	Cluster 2
A15 (6,11)	6.705	2.343	4.014	Cluster 2

Working of K-means algorithm: Example

Now, we will calculate the new centroid for each cluster for the third iteration.

1. In cluster 1, we have 5 points i.e. A2 (2,6), A5 (6,4), A6 (1,2), A10 (7,5), and A12 (4,6). To calculate the new centroid for cluster 1, we will find the mean of the x and y coordinates of each point in the cluster. Hence, the new centroid for cluster 1 is
2. In cluster 2, we have 7 points i.e. A1 (2,10), A4 (6,9), A7 (5,10) , A8 (4,9), A13 (3,10), A14 (3,8), and A15 (6,11). Hence, the new centroid for cluster 2 is
3. In cluster 3, we have 3 points i.e. A3 (11,11), A9 (10,12), and A11 (9,11). Hence, the new centroid for cluster 3 is

Working of K-means algorithm: Example

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2. In cluster 2, we have 7 points i.e. A1 (2,10), A4 (6,9), A7 (5,10) , A8 (4,9), A13 (3,10), A14 (3,8), and A15 (6,11). Hence, the new centroid for cluster 2 is (4.143, 9.571)
3. In cluster 3, we have 3 points i.e. A3 (11,11), A9 (10,12), and A11 (9,11). Hence, the new centroid for cluster 3 is (10, 11.333).

Working of K-means algorithm: Example

Here, you can observe that no point has changed its cluster compared to the previous iteration. Due to this, the centroid also remains constant. Therefore, we will say that the clusters have been stabilized. Hence, the clusters obtained after the third iteration are the final clusters made from the given dataset. If we plot the clusters on a graph, the graph looks like as follows.

How to Determine the Optimal K for K-Means?

1. The Elbow Method
2. The Silhouette Method (Assignment)

How to Determine the Optimal K for K-Means?

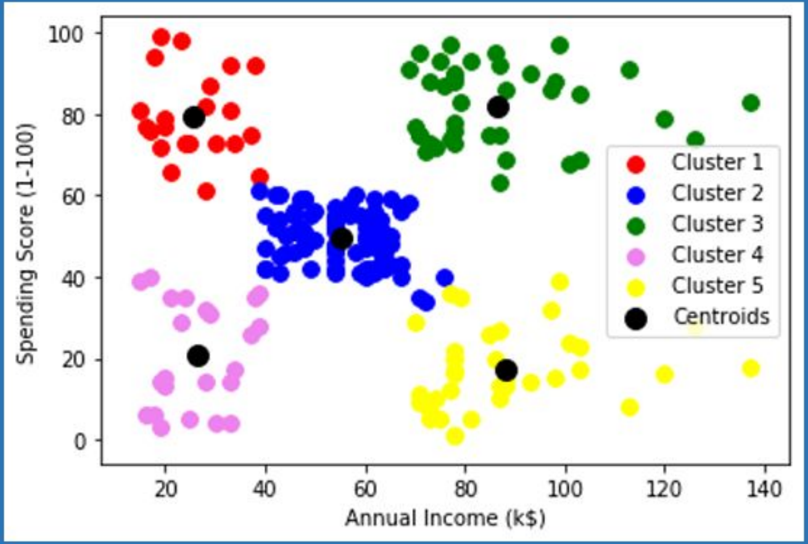
The Elbow Method: This is probably the most well-known method for determining the optimal number of clusters. It is also a bit naive in its approach.

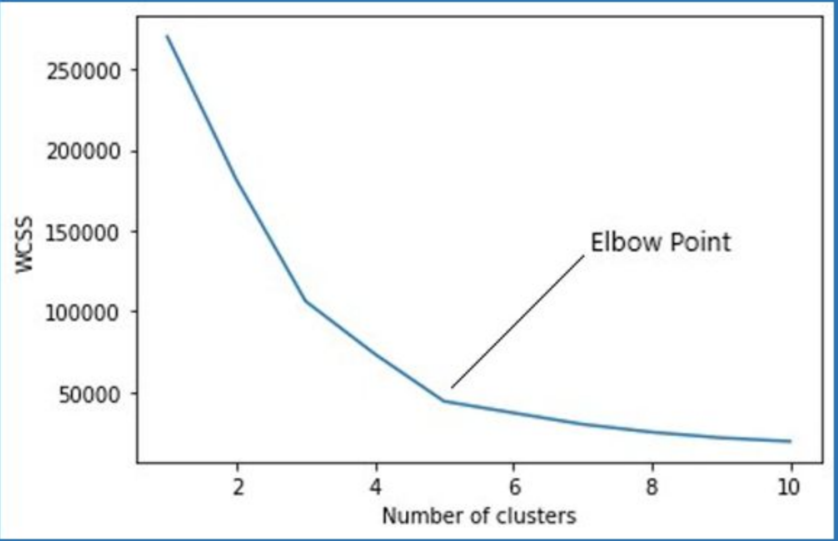
Calculate the Within-Cluster-Sum of Squared Errors (WSS) for different values of k, and choose the k for which WSS becomes first starts to diminish. In the plot of WSS-versus-k, this is visible as an elbow.

How to Determine the Optimal K for K-Means?

$$\mathrm{WCSS}(k) = \sum_{j=1}^{k} \sum_{\mathbf{x}_i \in \text{cluster } j} \|\mathbf{x}_i - \bar{\mathbf{x}}_j\|^2,$$

where $\bar{\mathbf{x}}_j$ is the sample mean in cluster j





A Simple example showing the implementation of k-means algorithm (using K=2)

Individual	Variable 1	Variable 2
1	1.0	1.0
2	1.5	2.0
3	3.0	4.0
4	5.0	7.0
5	3.5	5.0
6	4.5	5.0
7	3.5	4.5

Step 1:

Initialization: Randomly we choose following two centroids (k=2) for two clusters.

In this case the 2 centroid are: m1=(1.0,1.0) and m2=(5.0,7.0).

Individual	Variable 1	Variable 2
1	1.0	1.0
2	1.5	2.0
3	3.0	4.0
4	5.0	7.0
5	3.5	5.0
6	4.5	5.0
7	3.5	4.5

	Individual	Mean Vector
Group 1	1	(1.0, 1.0)
Group 2	4	(5.0, 7.0)

Step 2:

- Thus, we obtain two clusters containing:
  {1,2,3} and {4,5,6,7}.
- Their new centroids are:

Individual	Centroid 1	Centroid 2
1	0	7.21
2 (1.5, 2.0)	1.12	6.10
3	3.61	3.61
4	7.21	0
5	4.72	2.5
6	5.31	2.06
7	4.30	2.92

$$m_1 = \left(\frac{1}{3}(1.0+1.5+3.0), \frac{1}{3}(1.0+2.0+4.0)\right) = (1.83, 2.33)$$

$$m_2 = \left(\frac{1}{4}(5.0+3.5+4.5+3.5), \frac{1}{4}(7.0+5.0+5.0+4.5)\right)$$

$$= (4.12, 5.38)$$

$$d(m_1, 2) = \sqrt{|1.0-1.5|^2 + |1.0-2.0|^2} = 1.12$$

$$d(m_2, 2) = \sqrt{|5.0-1.5|^2 + |7.0-2.0|^2} = 6.10$$

Step 3:

- Now using these centroids we compute the Euclidean distance of each object, as shown in table.

- Therefore, the new clusters are:
  {1,2} and {**3**,4,5,6,7}

- Next centroids are: m1=(1.25,1.5) and m2 = (3.9,5.1)

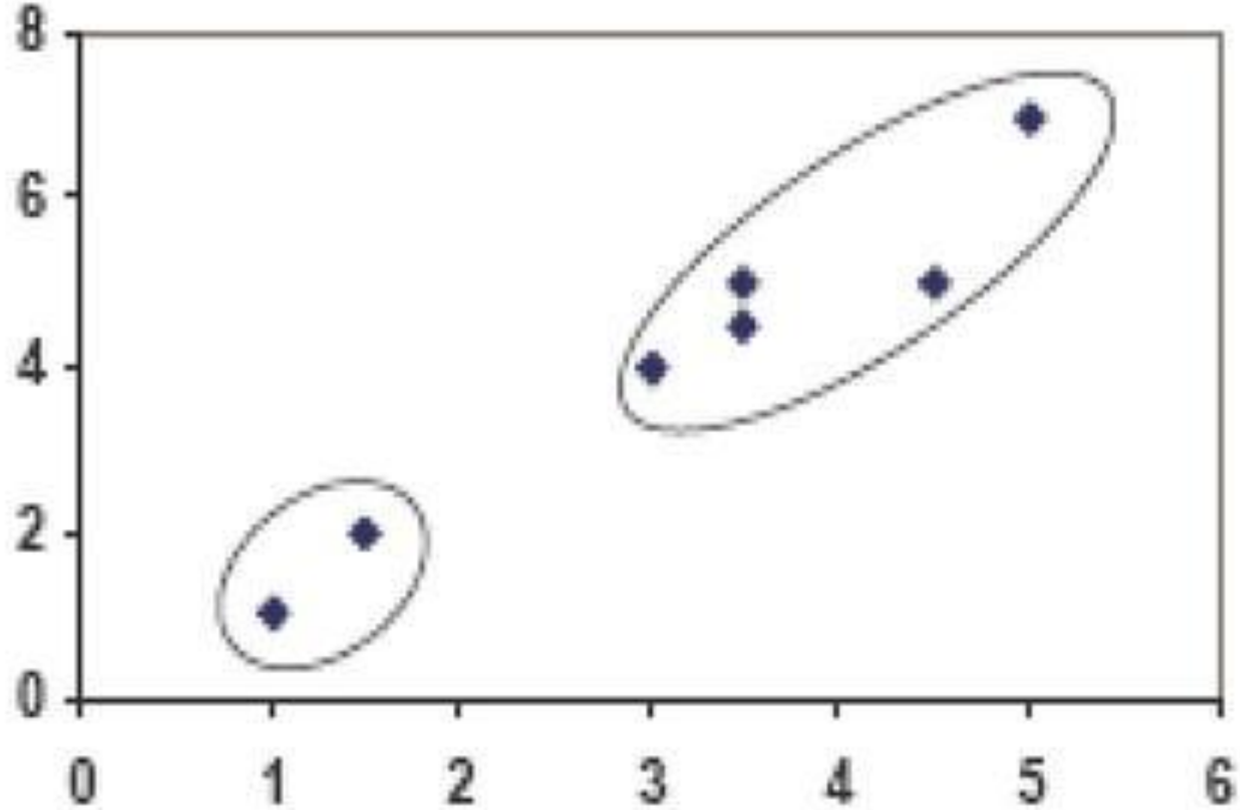
Individual	Centroid 1	Centroid 2
1	1.57	5.38
2	0.47	4.28
3	2.04	1.78
4	5.64	1.84
5	3.15	0.73
6	3.78	0.54
7	2.74	1.08

- Step 4 :
  The clusters obtained are:
  {1,2} and {3,4,5,6,7}

- Therefore, there is no change in the cluster.
- Thus, the algorithm comes to a halt here and final result consist of 2 clusters {1,2} and {3,4,5,6,7}.

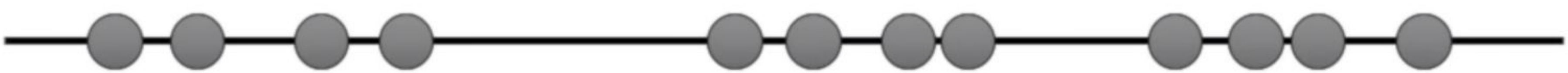
Individual	Centroid 1	Centroid 2
1	0.56	5.02
2	0. 56	3.92
3	3.05	1.42
4	6.66	2.20
5	4.16	0.41
6	4.78	0.61
7	3.75	0.72

PLOT

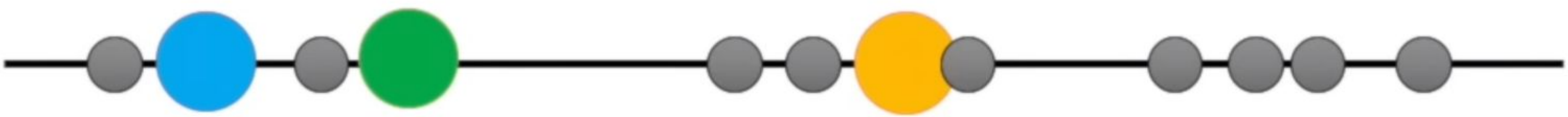


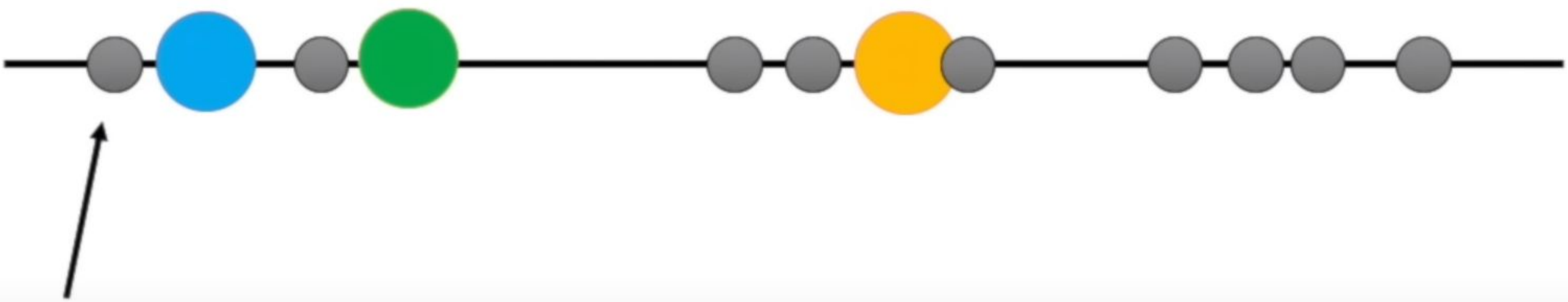
Step 1: Select the number of clusters you want to identify in your data. This is the "K" in "K-means clustering".

In this case, we'll select K=3. That is to say, we want to identify 3 clusters.

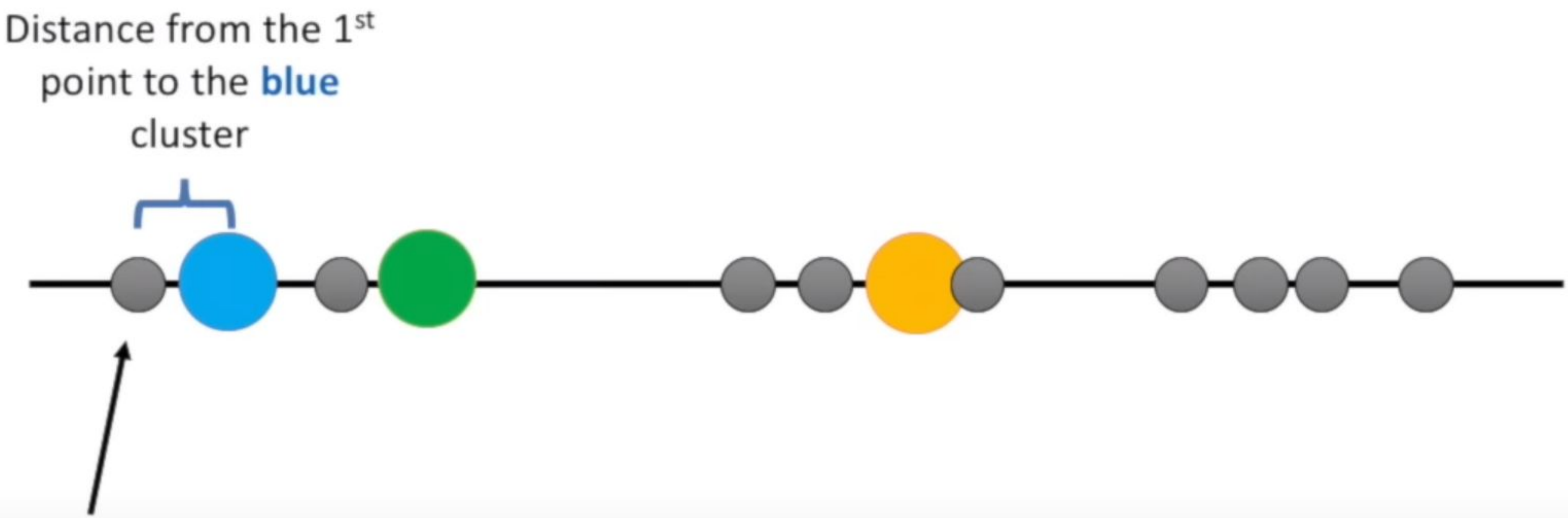


Step 2: Randomly select 3 distinct data points.

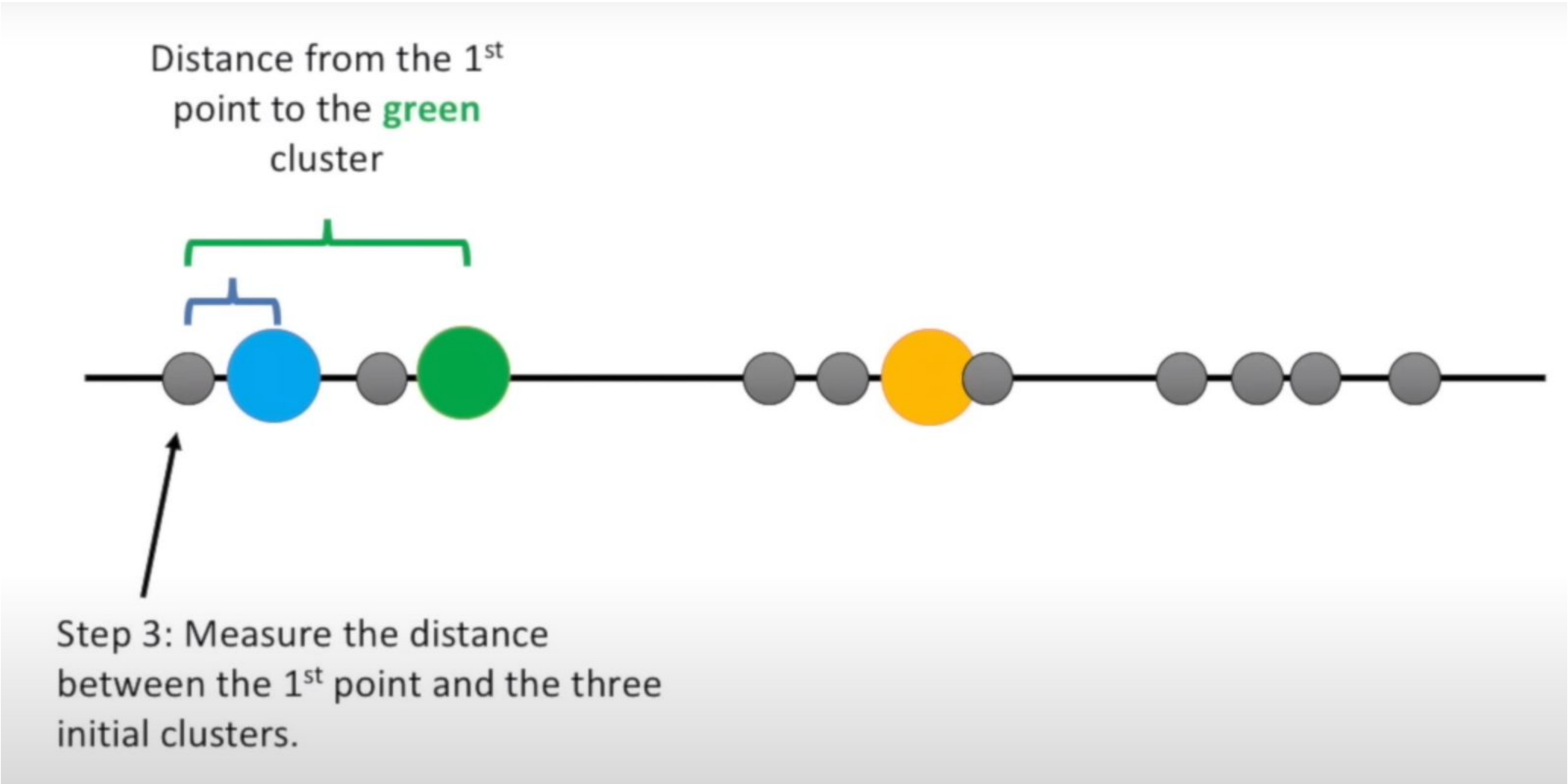




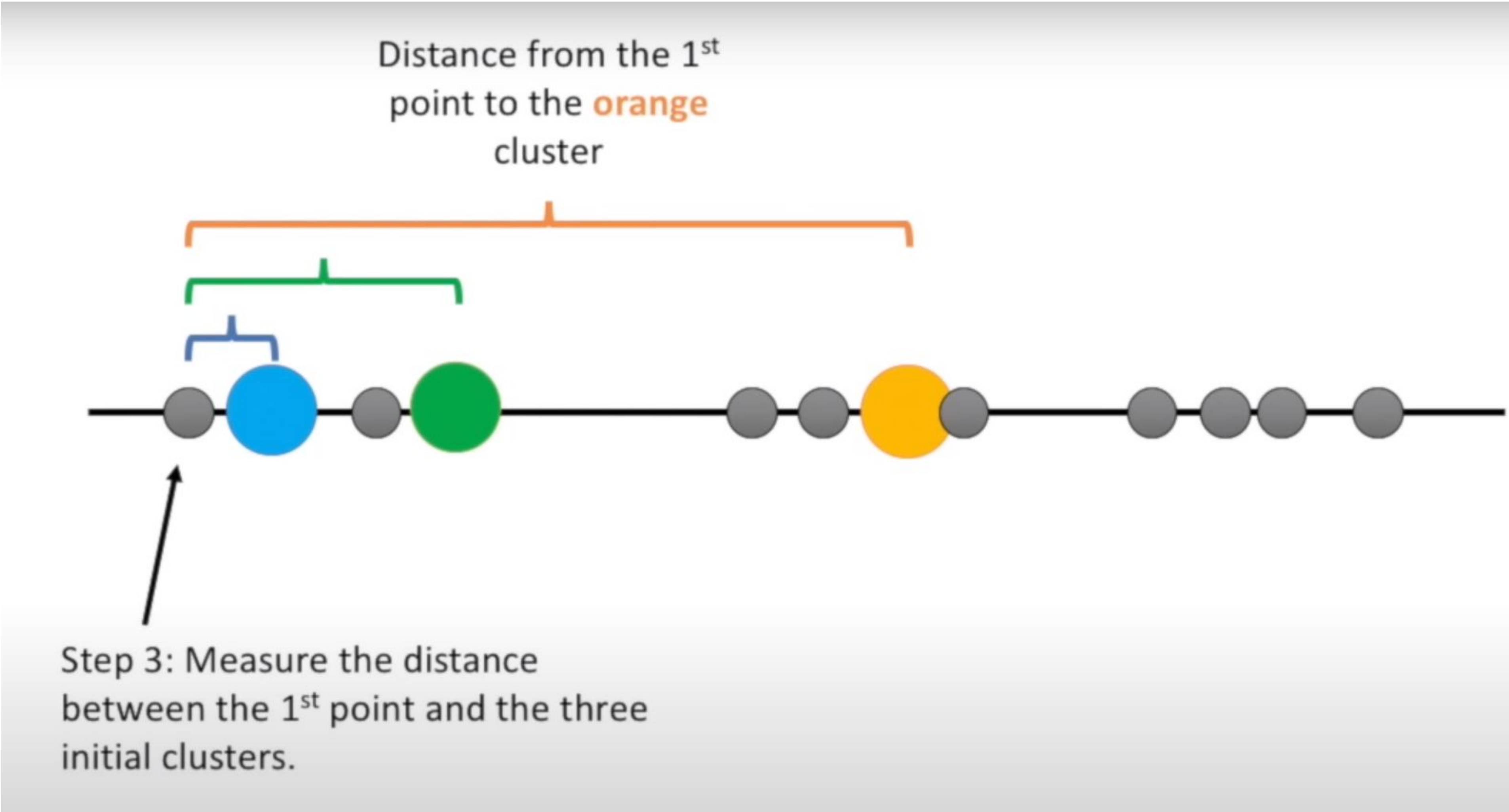
Step 3: Measure the distance between the 1st point and the three initial clusters.

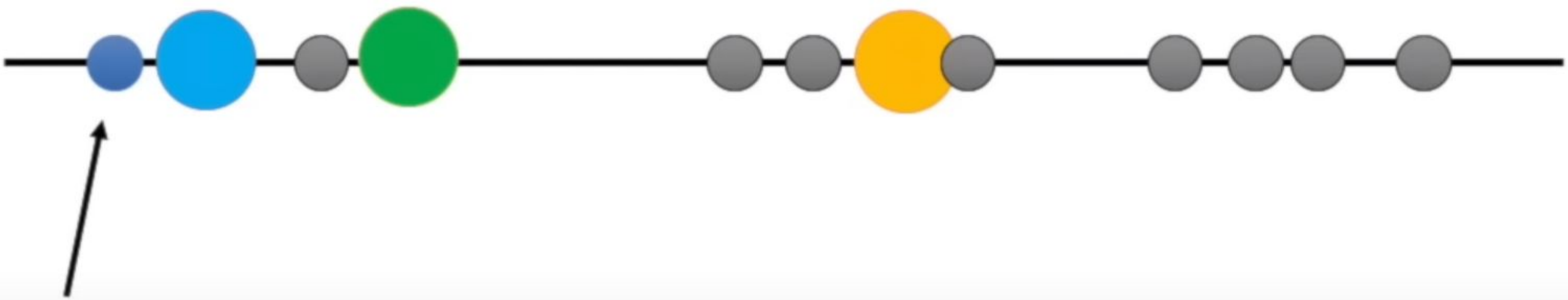


Step 3: Measure the distance between the 1st point and the three initial clusters.



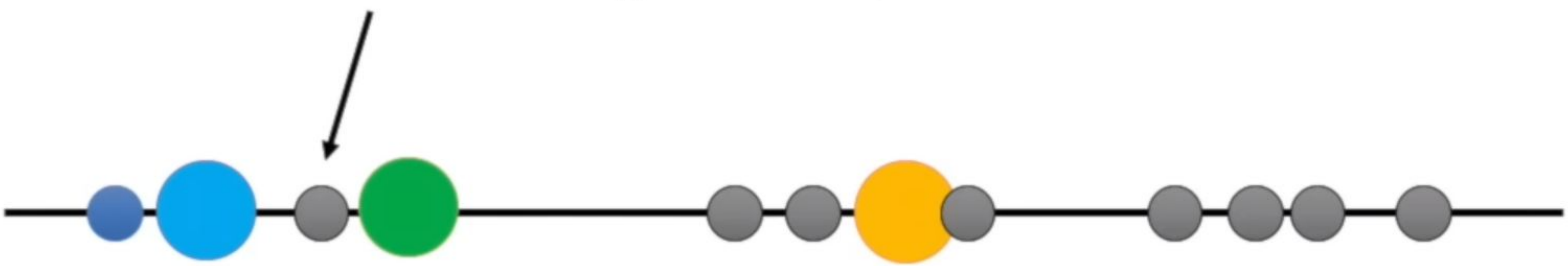
Distance from the 1st point to the green cluster
Step 3: Measure the distance between the 1st point and the three initial clusters.

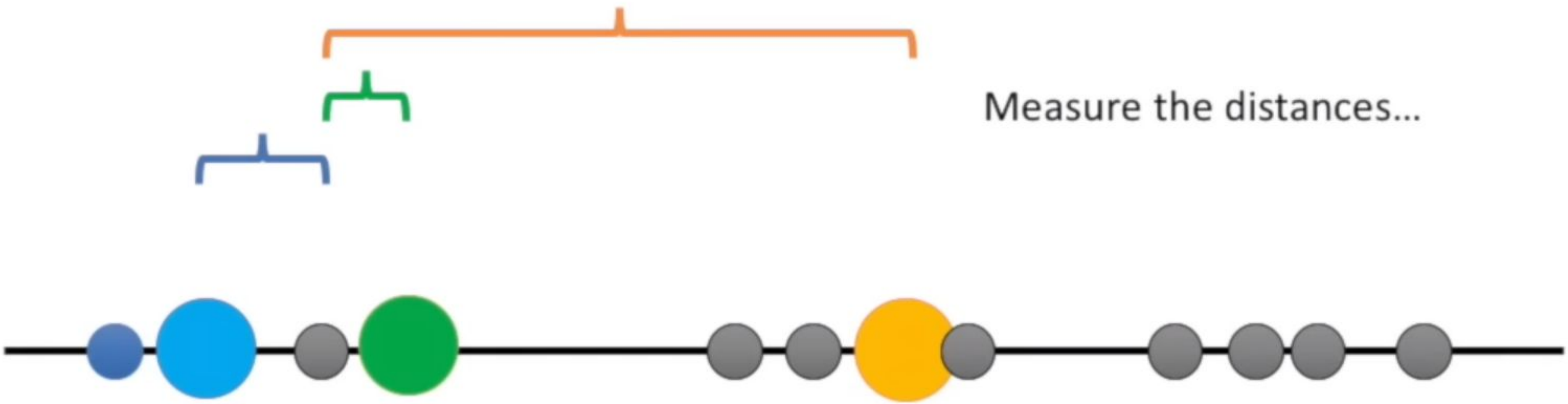

Distance from the 1st point to the orange cluster
Step 3: Measure the distance between the 1st point and the three initial clusters.

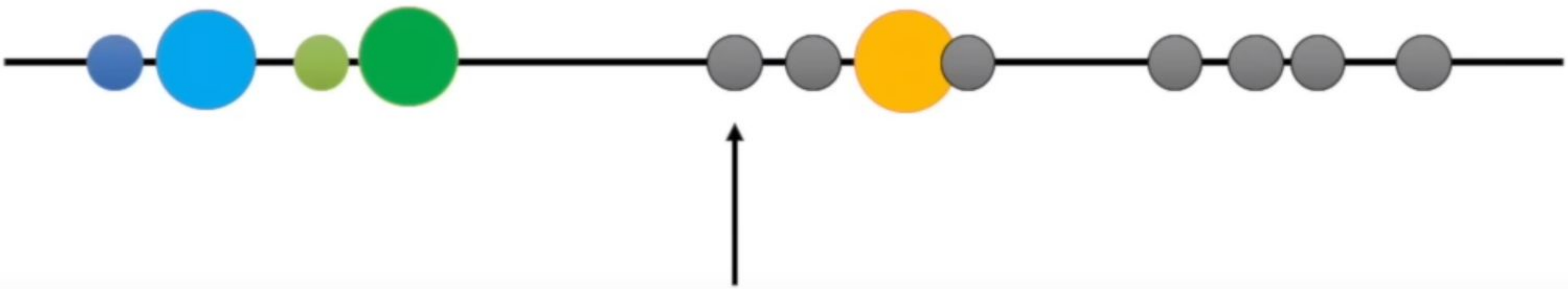


Step 4: Assign the 1st point to the nearest cluster. In this case, the nearest cluster is the **blue** cluster.

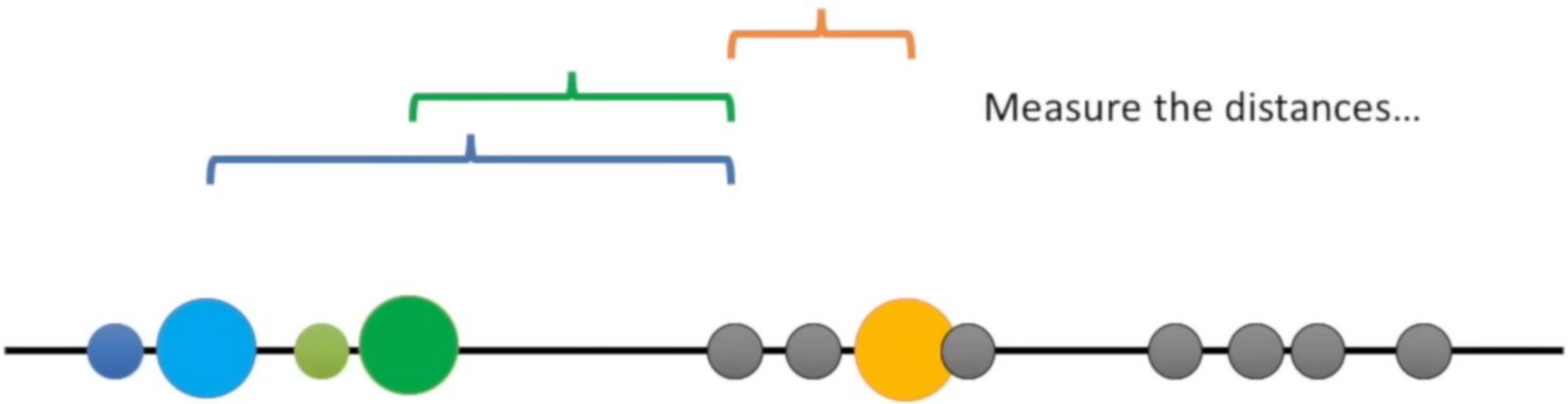
Now do the same thing for the next point.



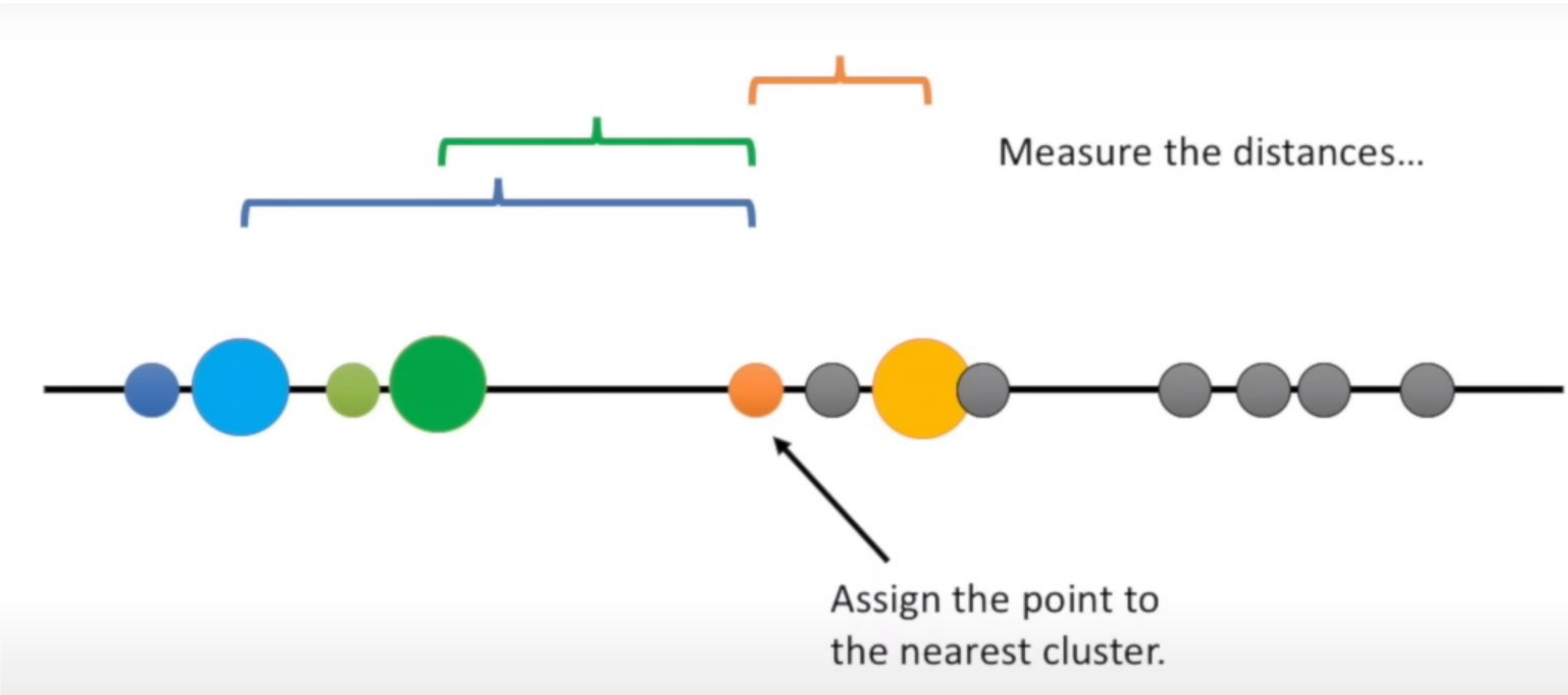

Measure the distances...

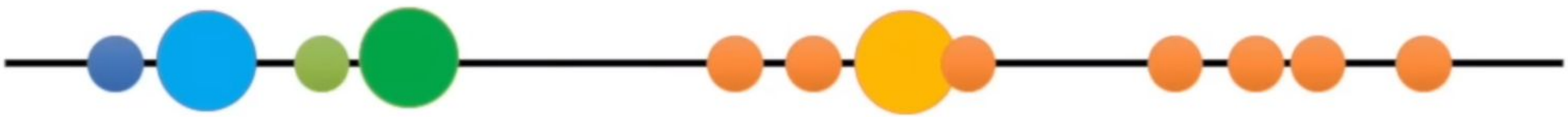


Now figure out which cluster the 3rd point belongs to.



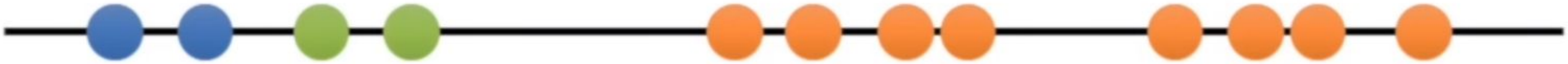
Measure the distances...


Measure the distances...
Assign the point to
the nearest cluster.

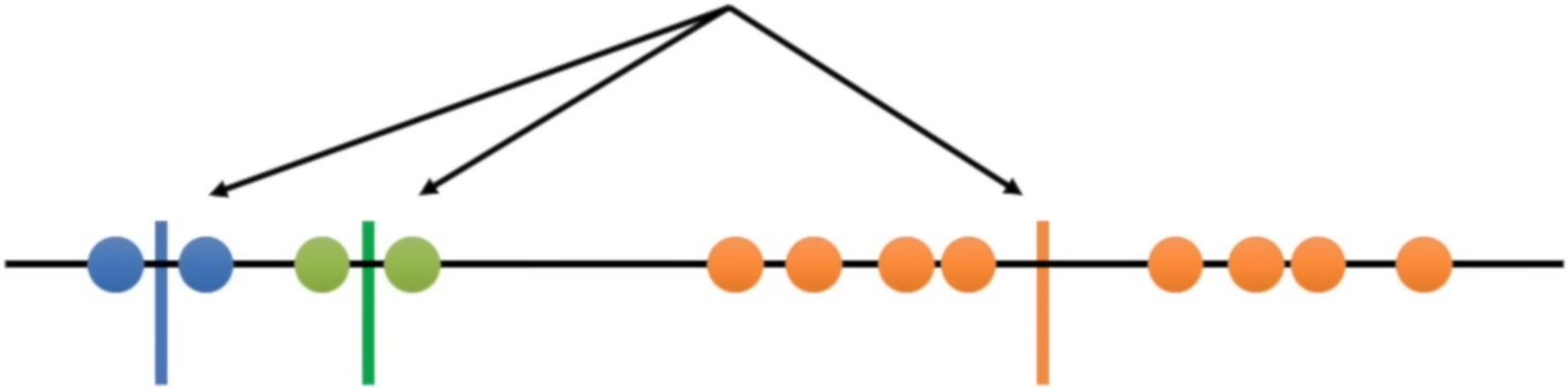


The rest of these points are closest to the **orange** cluster, so they'll go in that one, too.

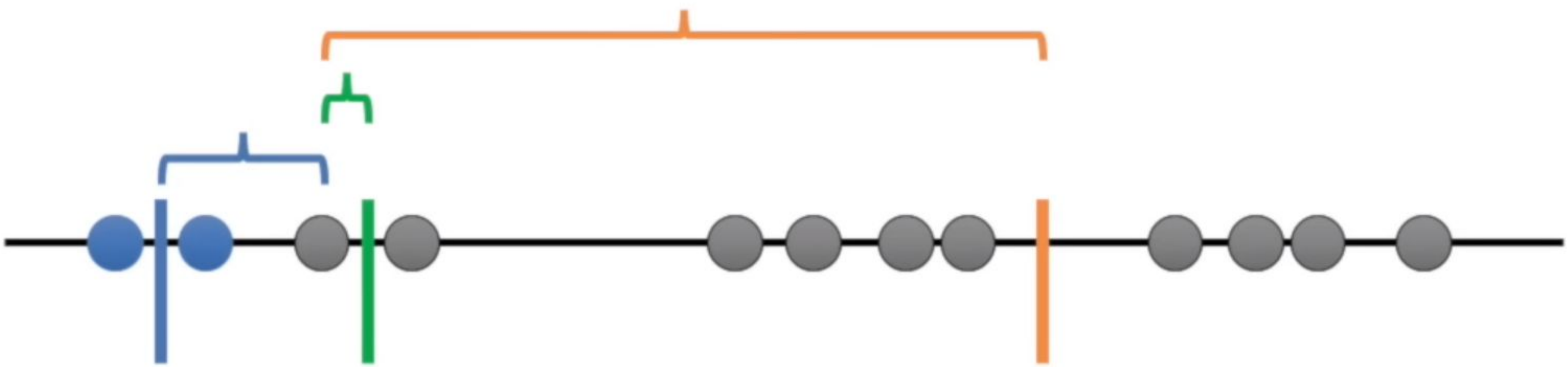
Now that all of the points are
in clusters, we go on to…



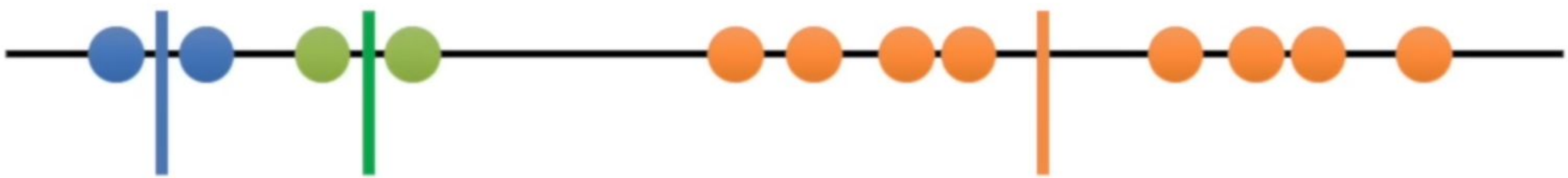
Step 5: calculate the mean of each cluster.



Then we repeat what we just
did (measure and cluster)
using the mean values.



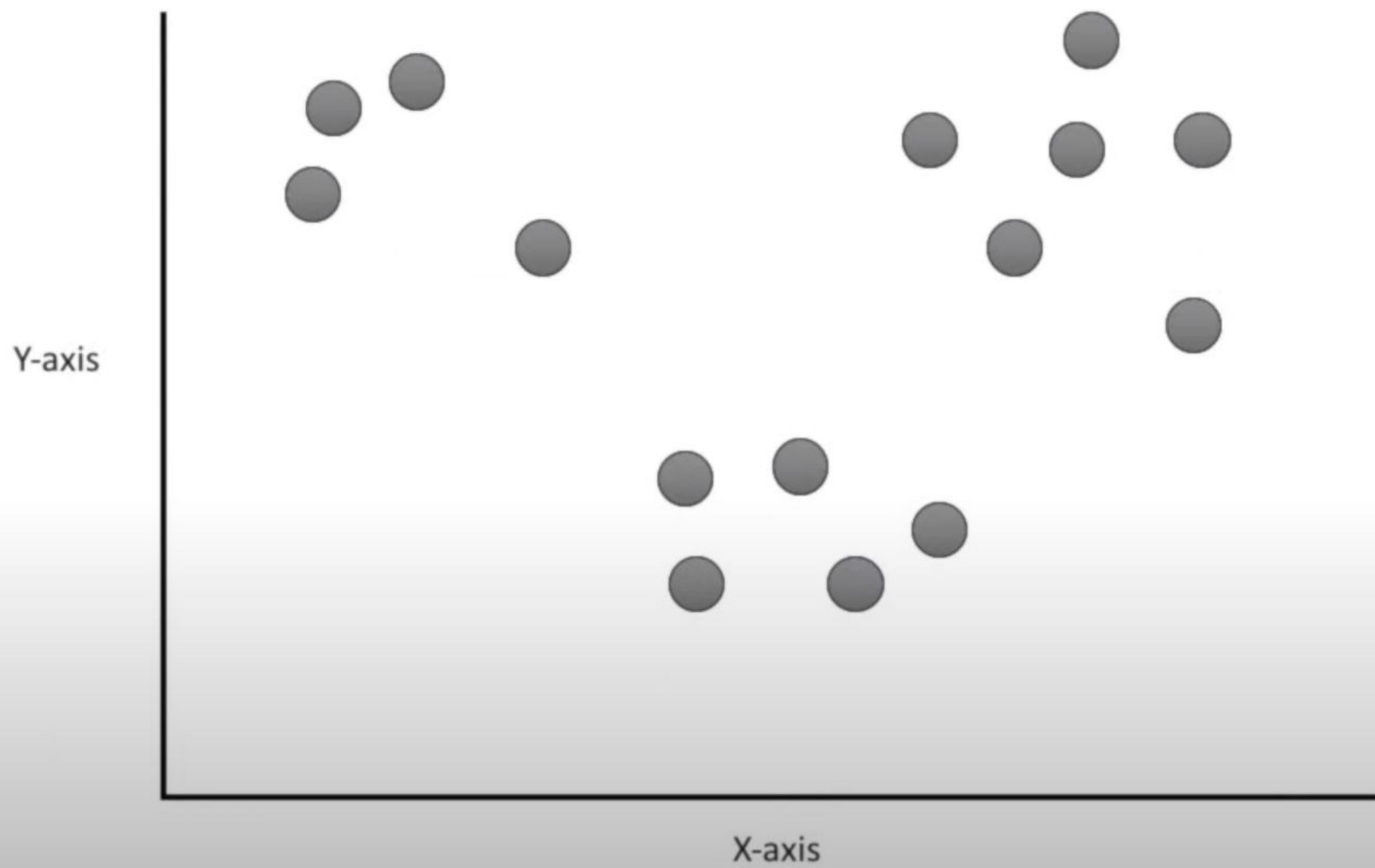
Since the clustering did not change at all during the last iteration, we're done…



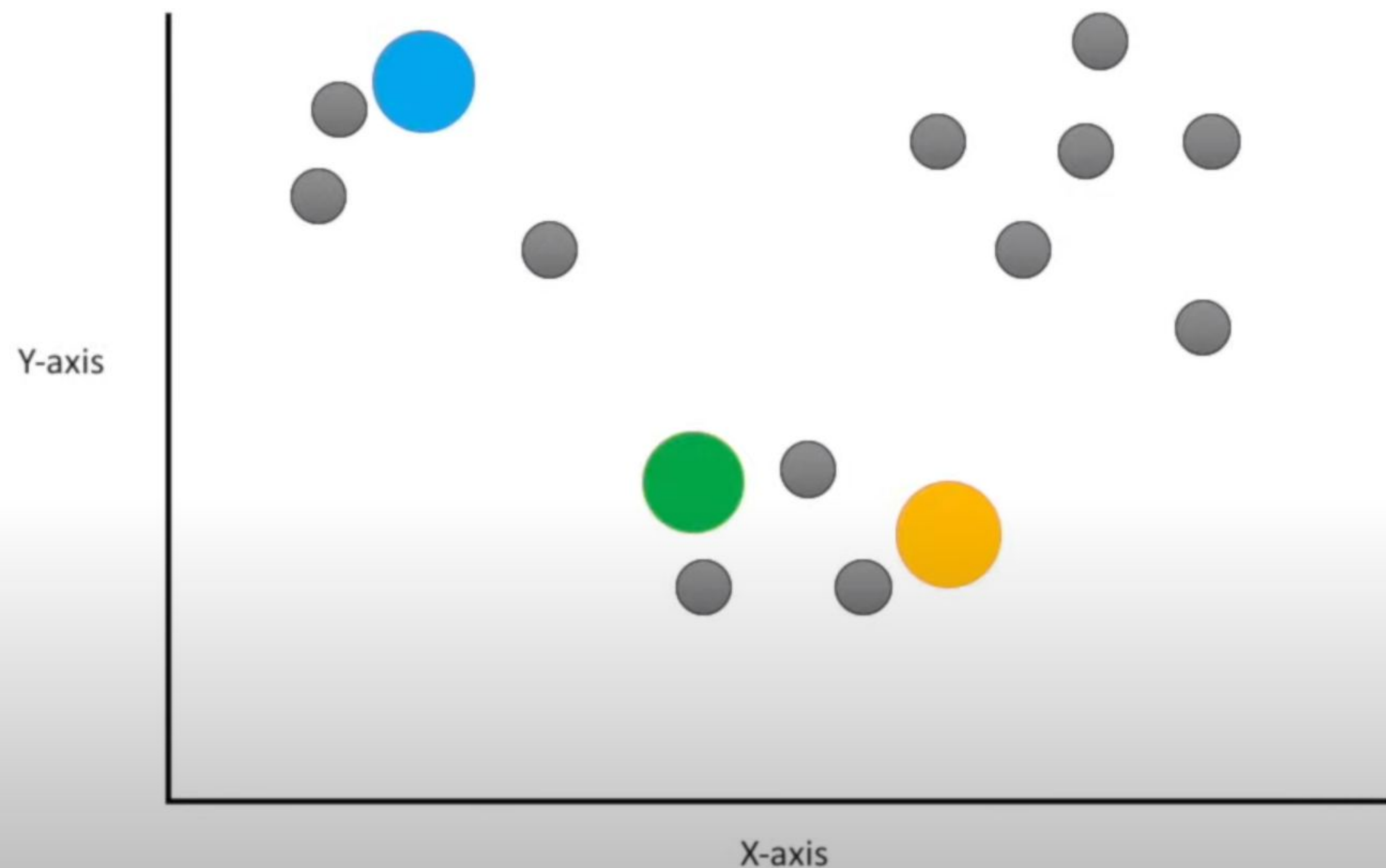
Question: How is K-means clustering different from hierarchical clustering?

K-means clustering specifically tries to put the data into the number of clusters you tell it to.

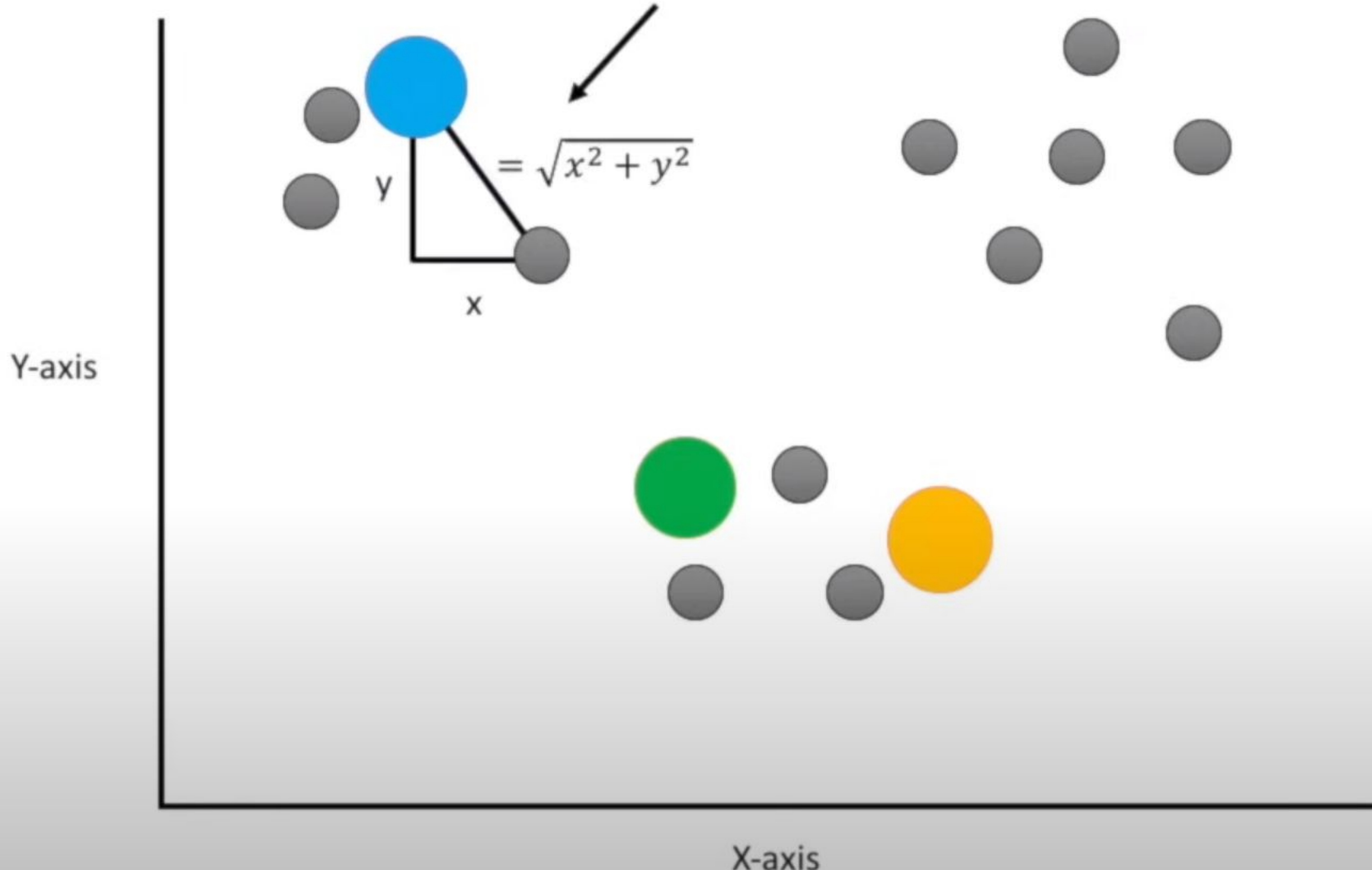
Question: What if our data isn't plotted on a number line?

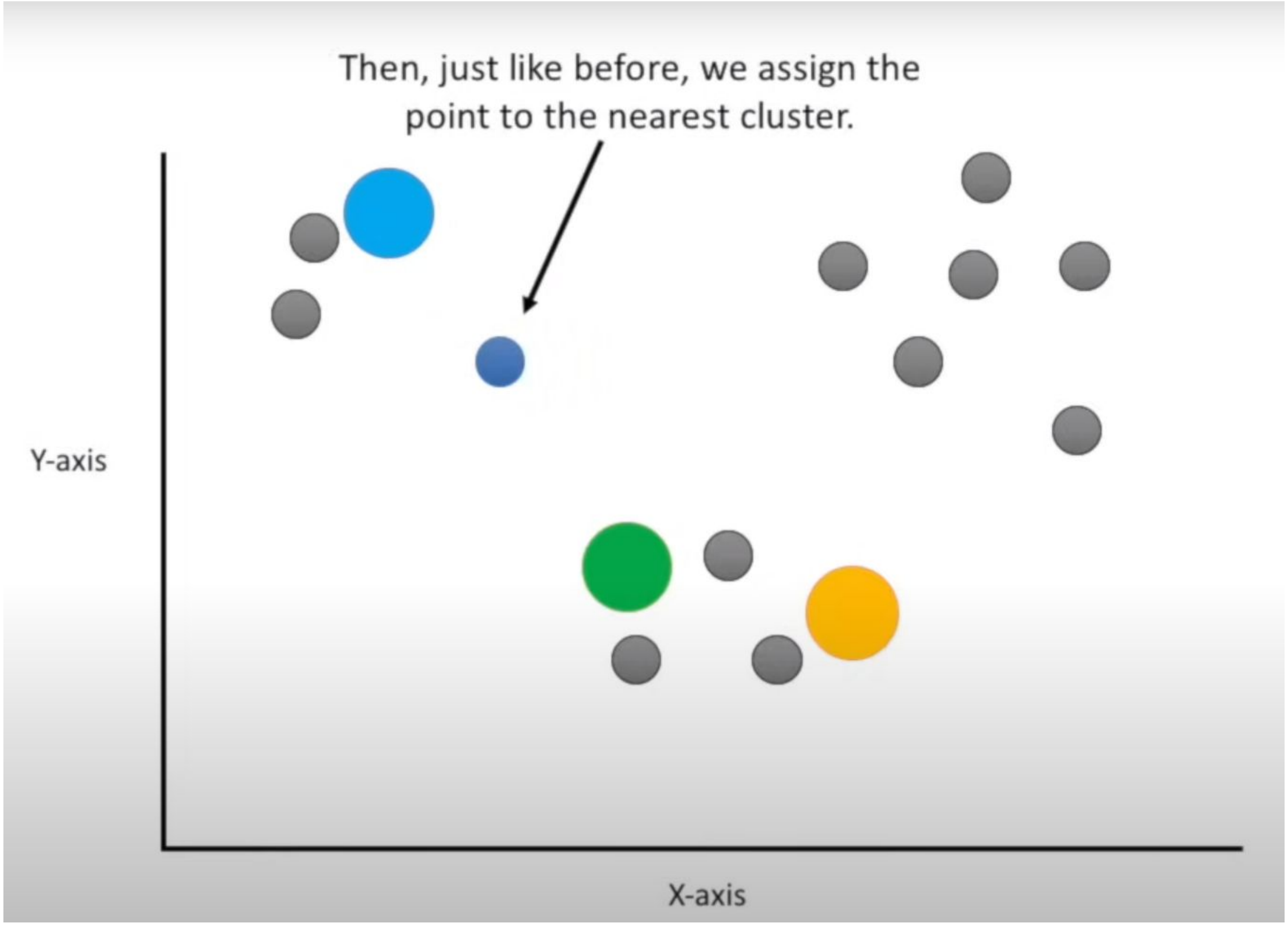


Just like before, you pick three random points...

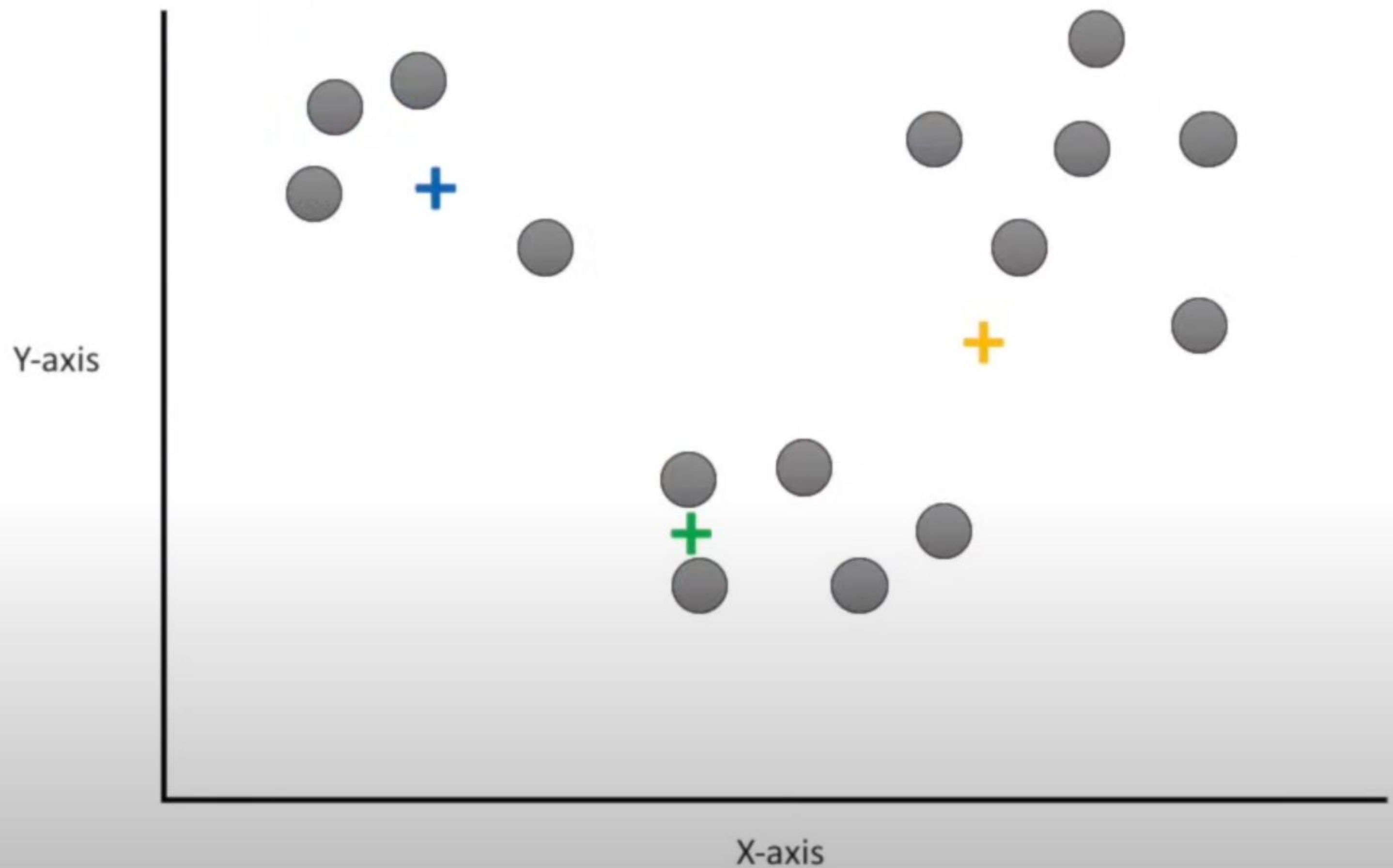


And we use the Euclidean distance. In 2 dimensions, the Euclidian distance is the same thing as the Pythagorean theorem.




Then, just like before, we assign the
point to the nearest cluster.
Y-axis
X-axis

And, just like before, we then calculate the center of each cluster and recluster...



Weaknesses of K-Mean Clustering

1. When the numbers of data are not so many, initial grouping will determine the cluster significantly.
2. The number of cluster, K, must be determined before hand. Its disadvantage is that it does not yield the same result with each run, since the resulting clusters depend on the initial random assignments.
3. We never know the real cluster, using the same data, because if it is inputted in a different order it may produce different cluster if the number of data is few.
4. It is sensitive to initial condition. Different initial condition may produce different result of cluster. The algorithm may be trapped in the *local optimum*.

Applications of K-Mean Clustering

- It is relatively *efficient and fast.* It computes result at **O(tkn),** where n is number of objects or points, k is number of clusters and t is number of iterations.
- k-means clustering can be applied to *machine learning or data mining*
- *Used on acoustic data in speech understanding to convert waveforms into one of k categories (known as Vector Quantization or Image Segmentation).*
- *Also used for choosing color palettes on old fashioned graphical display devices and Image Quantization.*

CONCLUSION

- *K-means algorithm is* useful for undirected knowledge discovery and is relatively simple. K-means has found wide spread usage in lot of fields, ranging from unsupervised learning of neural network, Pattern recognitions, Classification analysis, Artificial intelligence, image processing, machine vision, and many others.

References

- Tutorial - Tutorial with introduction of Clustering Algorithms (k-means, fuzzy-c-means, hierarchical, mixture of gaussians) + some interactive demos (java applets).
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- www.wikipedia.com